

# Chapter Five

## Introduction to Computer Networks



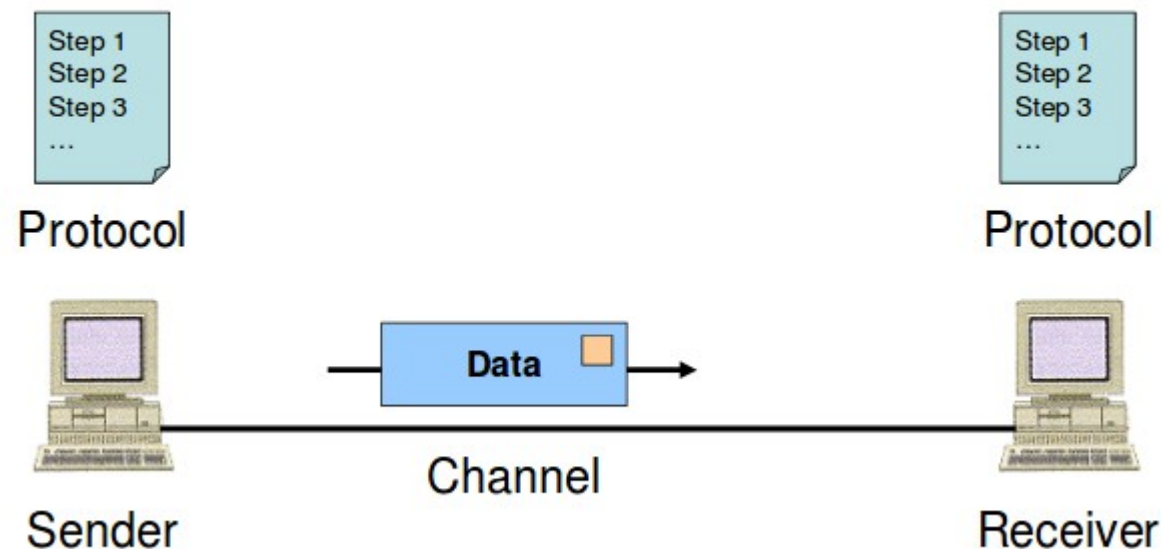
# Contents



- Basics of Data Communication
- Data Transmission
- Communication Mode
- Computer Networks and Its Application
- Types of Networks
- Network Models and Topologies
- Transmission Media
- The Internet and Web Concept

# Basics of Data communication

- The term **telecommunication** means communication at a distance
- The word **data** refers to information presented in whatever form is agreed upon by the parties **creating** and **using the data**
- **Data communications** are the exchange of data between two devices via some form of transmission medium such as a **wire cable**.



# Basics of Data communication (Cont.)

The **effectiveness** of a data communication system depends on four fundamental characteristics:

1. **Delivery**: The system must **deliver data** to the correct destination.
  - Data must be received by the **intended device** or user and only by that device or user.
2. **Accuracy**: The system must deliver the data accurately.

# Basics of Data communication (Cont.)

3. **Timeliness**: The system must deliver data in a timely manner.

- Data delivered late are useless.

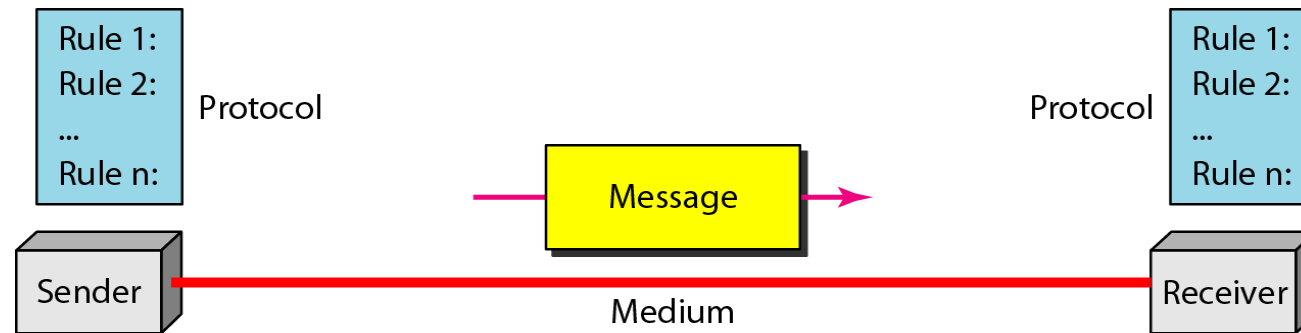
4. **Jitter**: Jitter refers to the variation in the packet arrival time.

- It is an uneven delay in the delivery of audio or video packets.

**Example**: For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

# Data Communication Component

- The following are the basic components for working a communication system.



1. Transmitter (Sender),
2. Receiver,
3. Medium,
4. Message, and
5. Protocol

# Data Communication Component (Cont.)

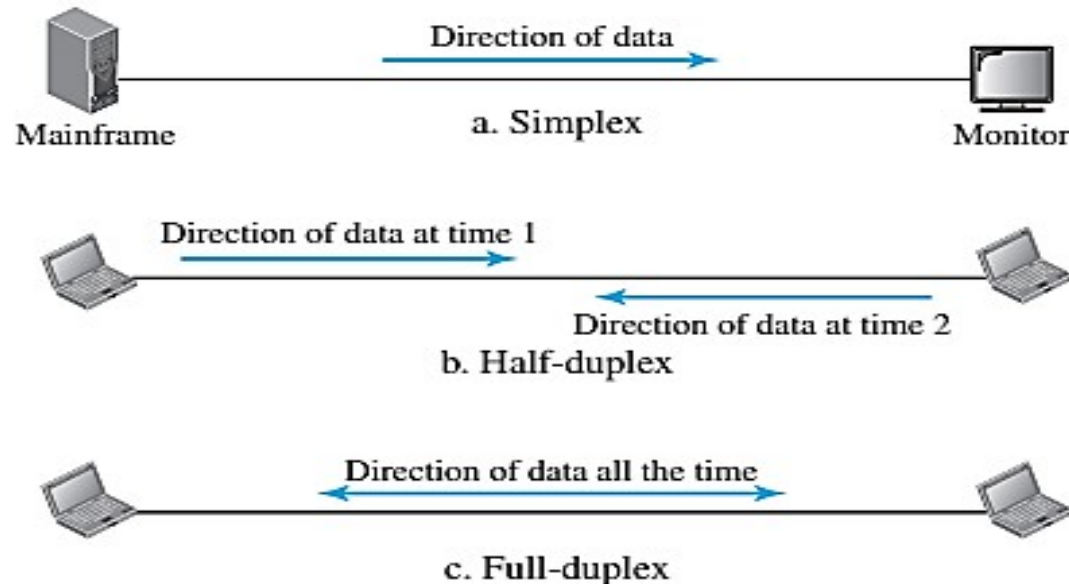
1. **Transmitter (Sender)**: is the device that sends the message.
  - It can be a computer, workstation, telephone handset, video camera, and so on.
2. **Receiver**: is the device that receives the message.
  - It can be a computer, workstation, telephone handset, television, and so on.
3. **Medium**: The transmission medium is the physical path by which a message travels from sender to receiver.
  - It can consist of twisted pair wire, coaxial cable, fiber-optic cable, laser or radio waves (terrestrial or satellite microwave).

# Data Communication Component (Cont.)

4. **Message**: The message is the transmission (data) to be communicated.
- It can consist of **text**, **number**, **pictures**, **sound**, or **video** or any combination of these.
5. **Protocol**: A protocol is a set of rules that governs data communication.
- It represents an agreement between the **communicating devices**.
  - Without a protocol, two devices may be **connected** but **not communicating**.
  - Just as a person speaking **German** cannot be understood by a person who speaks only **Japanese**.

# Communication mode

- Communication mode refers to the how the data transfer between two devices.
- Communication between two devices can be **simplex**, **half-duplex**, or **full-duplex**.



## i. Simplex

- In this mode the communication is **unidirectional**, as on a one-way street.
- Only one of the two devices on a link can **transmit**; the other can only receive (see Figure 2 a).
- **Keyboards** and **traditional monitors** are examples of simplex devices.
- The **keyboard can only introduce input**; the monitor can only accept output.
- The simplex mode can use the entire capacity of the channel to send data **in one direction**.

## ii. Half-Duplex

- Each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa (see Figure 2 b).
- The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.
- The entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

### iii. Full-Duplex

- Both stations can transmit and receive simultaneously.
- Is like a two-way street with traffic flowing in both directions at the same time.
- Signals going in one direction share the capacity of the link with signals going in the other direction.
- This sharing can occur in two ways:
  - i. Either the link must contain two physically separate transmission paths, one for sending and the other for receiving;
  - ii. Or the capacity of the channel is divided between signals traveling in both directions.

### iii. Full-Duplex (Cont.)

- One common example of full-duplex communication is the **telephone network**.
- When two people are communicating by a telephone line, both can **talk** and **listen** at the same time.



# Computer Networks

- A computer network is the **interconnection** of various **computer systems** located at different places.
- In a computer network, two or more computers are linked together with a medium and data communication devices for the purpose of **communicating data** and **sharing resources**.
- The computer that provides resources to other computers on a network is known as a **server**.
- In the network, the individual computers, which access shared network resources, are known as **nodes**.

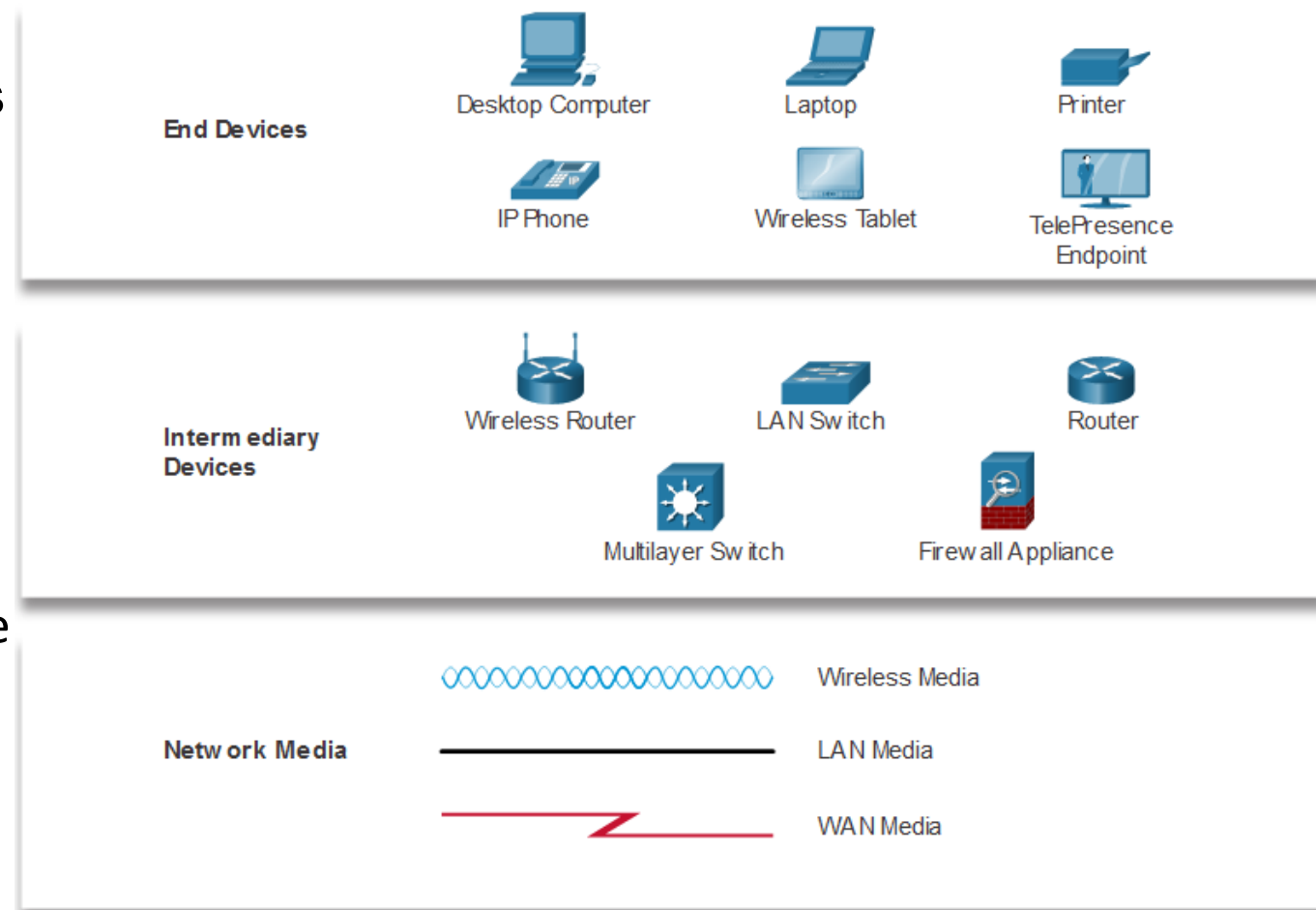
# Network Representations

Network diagrams, often called **topology diagrams**, use symbols to represent devices within the network.

Important terms to know include:

- Network Interface Card (NIC)
- Physical Port
- Interface

**Note:** Often, the terms port and interface are used interchangeably



# Application of Computer Networks



Some application of computer networks

- File sharing
- Printer / peripheral sharing
- Internet connection sharing
- Multi-player games

# Types of Computer Networks

Network infrastructures vary **greatly** in terms of:

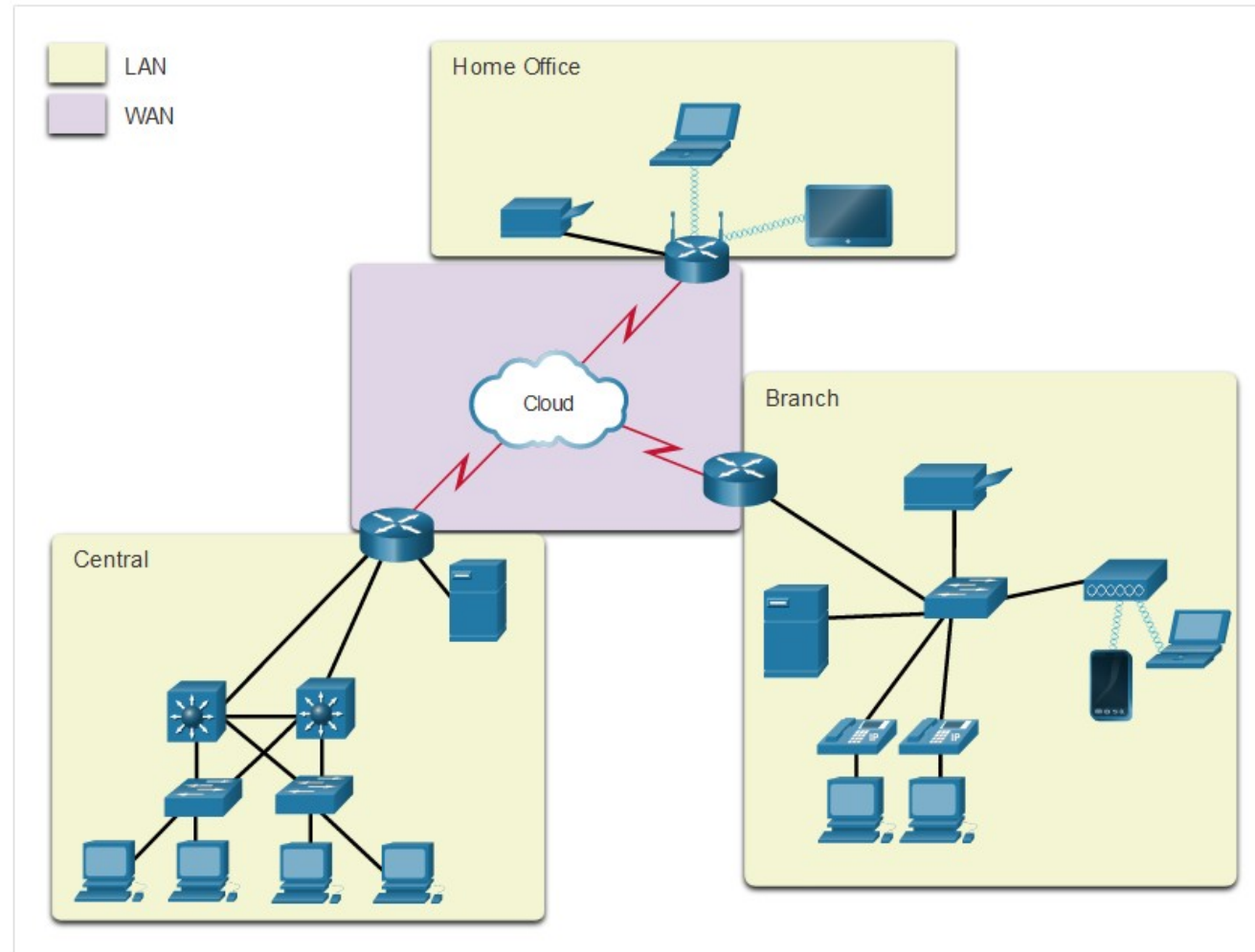
- Size of the **area** covered
- Number of **users** connected
- **Number** and **types** of services available
- Area of **responsibility**

From an **end user's point of view** there are three basic types:

- Local Area Network
- Wide Area Network
- Metropolitan Area Network

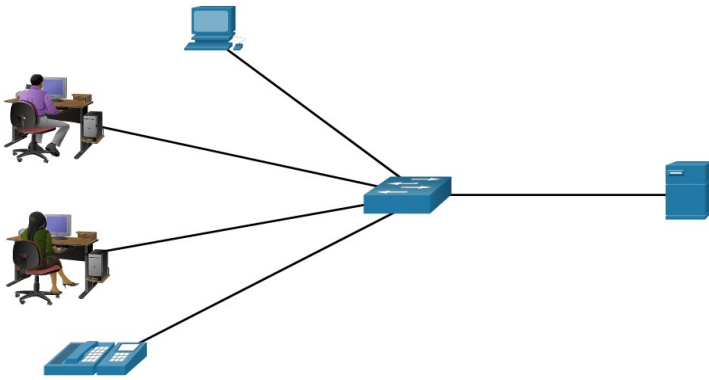
**Two most common types of networks:**

- Local Area Network (**LAN**) and Wide Area Network (**WAN**).

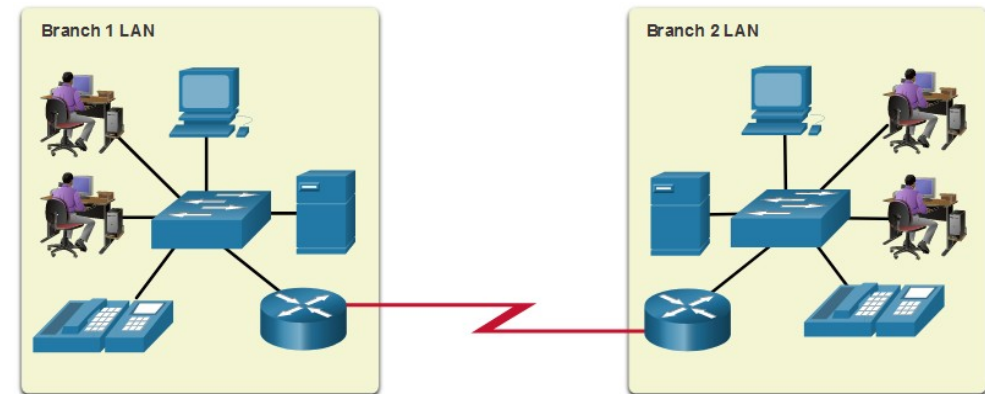


# Types of Computer Networks (Cont.)

- A LAN is a network infrastructure that spans a **small geographical area**.



- A WAN is a network infrastructure that spans a **wide geographical area**.

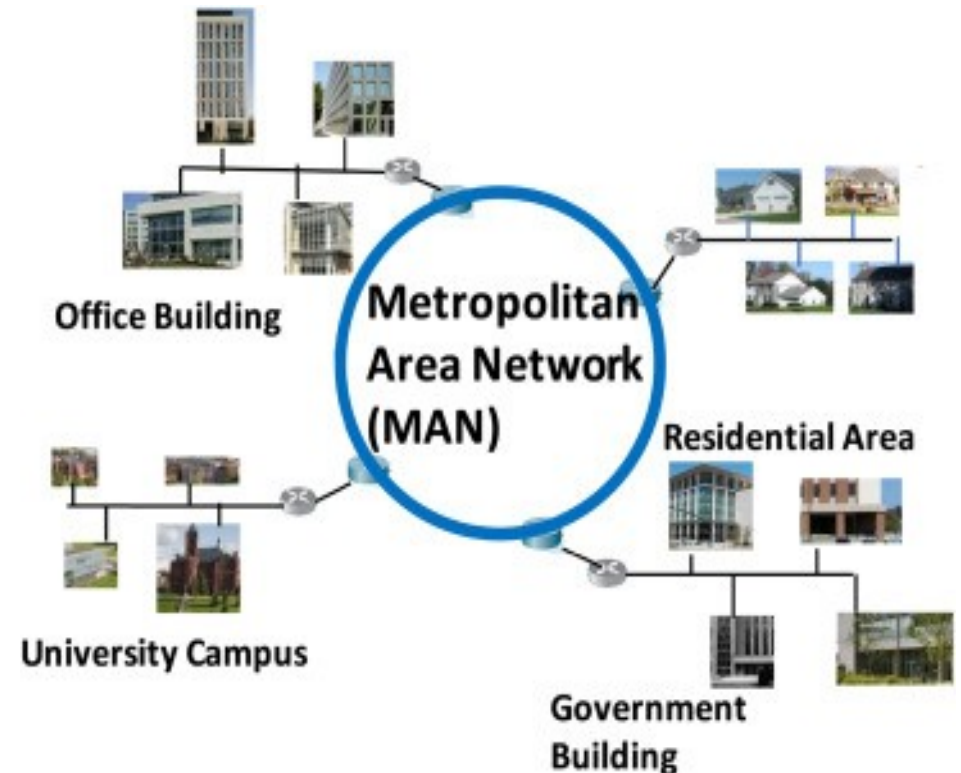


| LAN                                                                 | WAN                                                              |
|---------------------------------------------------------------------|------------------------------------------------------------------|
| Interconnect end devices in a <b>limited area</b> .                 | <b>Interconnect LANs</b> over wide geographical areas.           |
| Administered by a <b>single organization</b> or <b>individual</b> . | Typically administered by one or more <b>service providers</b> . |
| Provide <b>high-speed bandwidth</b> to internal devices.            | Typically provide <b>slower speed</b> links between LANs.        |

# Types of Computer Networks (Cont.)

## MAN

- It is in between **LAN** & **WAN** technology that covers the entire city.
- It uses similar technology as **LAN**.
- It can be a single network such as **cable TV network**, or a measure of connecting a number of LAN's on a large network so that resources can be shared LAN to LAN as well as device to device.

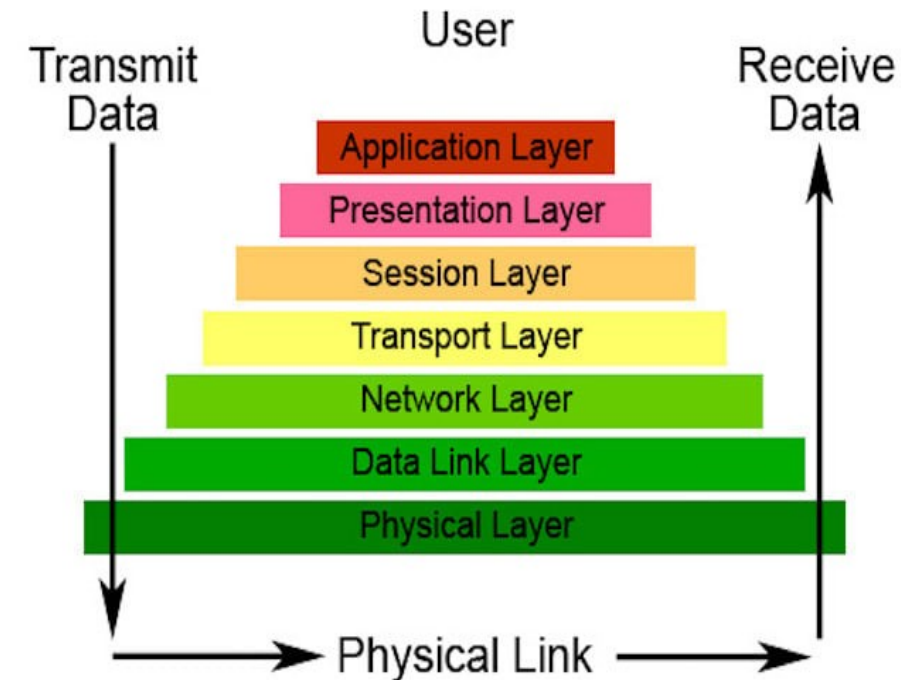


# Network Models

## The OSI Model

- ❑ The OSI stands for **Open System Interconnection**, which was developed in the 1980s.
- ❑ It is a conceptual model used for **network communication**.
- ❑ It is not implemented **entirely**, but it is still referenced today.
- ❑ The model consists of seven layers, and **each layer is connected** to the other.

## The Seven Layers of OSI



# Network Models (Cont.)

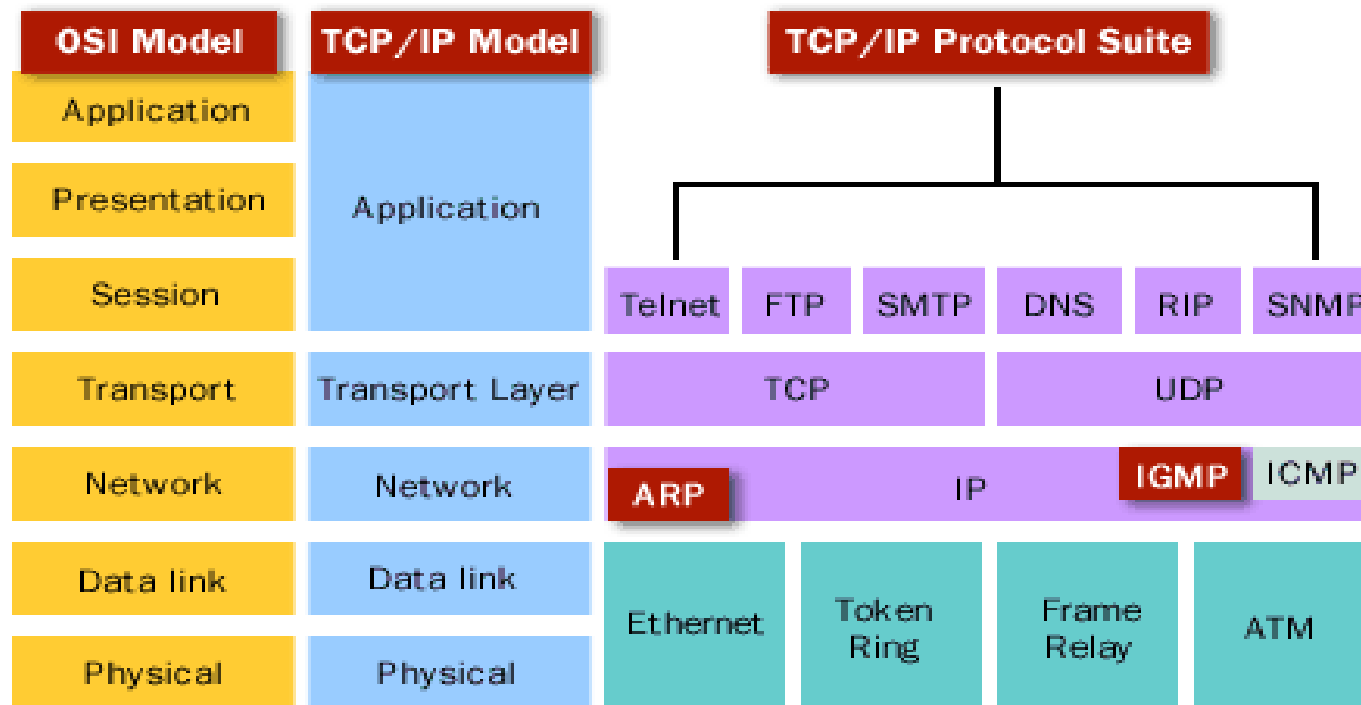
- ❑ The data moves down the **OSI model**, and each layer adds additional information.
- ❑ The data moves down until it reaches the last layer of the **OSI model**.
- ❑ When the data is received at the **last layer** of the OSI model, then the data is transmitted over the network.
- ❑ Once the data is reached on the other side, then the process will **get reversed**.

## TCP/IP Model

- ❑ The TCP model stands for **Transmission Control Protocol**, whereas **IP** stands for Internet Protocol.
- ❑ A number of protocols that make the **internet** possibly comes under the **TCP/IP model**.

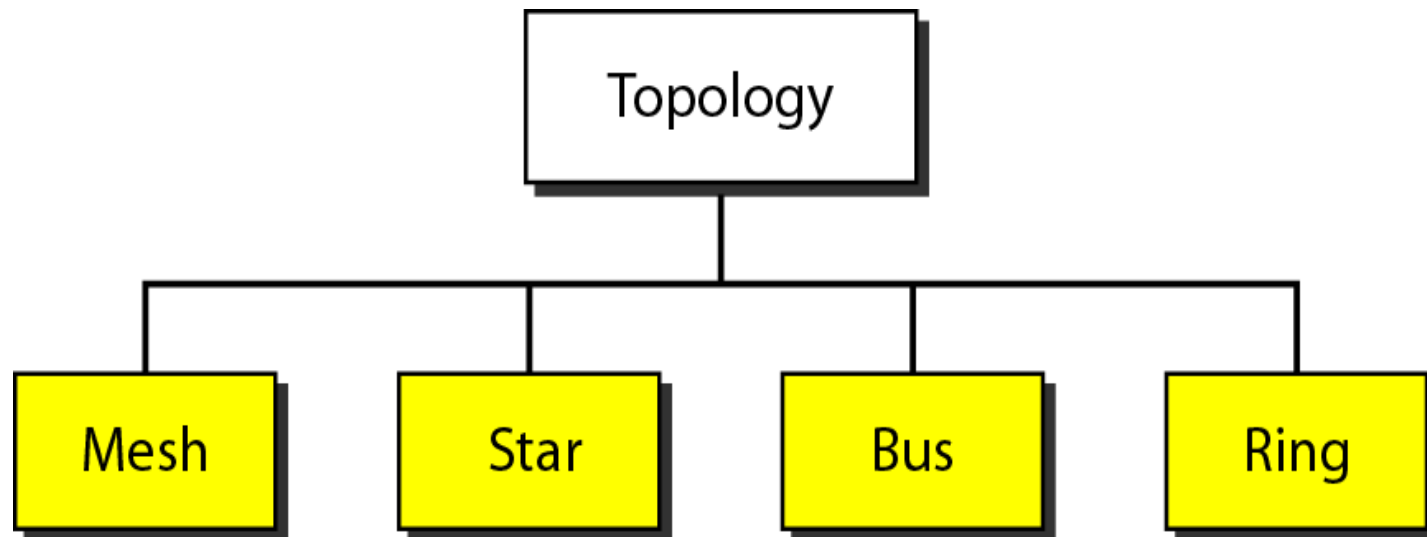
# TCP/IP Model (Cont.)

- Nowadays, we do not hear the name of the **TCP/IP model** much, we generally hear the name of the **IPv4** or **IPv6**, but it is still valid.
- This model consists of **4 layers**. Now, we will look at the diagrammatic representation of the **TCP/IP model**.



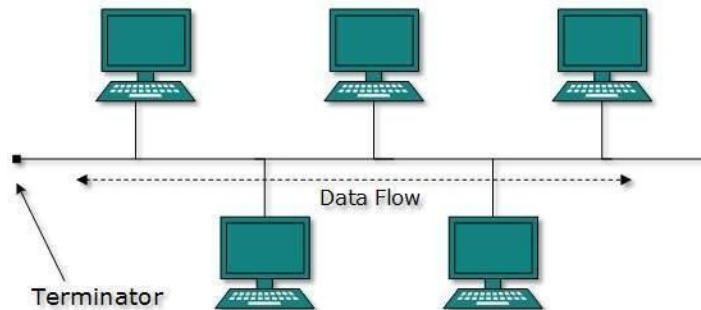
# Network Models (Cont.)

- ❑ Computers in a network have to be connected in some logical manner.
- ❑ The layout pattern of the interconnections between computers in a network is called **Network Topology**.
- ❑ Some of the network topologies is mentioned below.



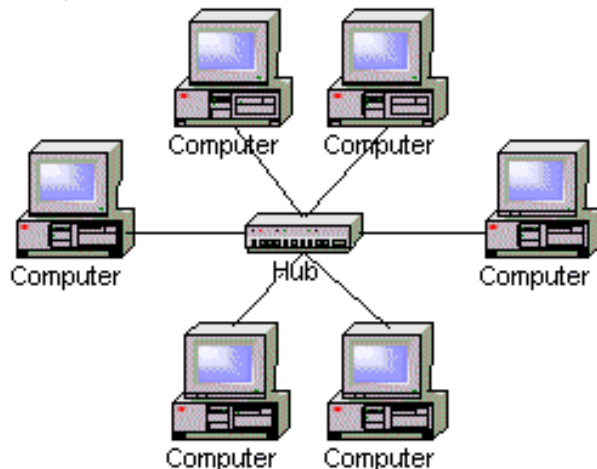
# Bus Topology

- ❑ Uses one main cable to **which all nodes** are directly connected.
- ❑ The main cable acts as a **backbone for the network**.
- ❑ One of the computers in the network typically acts as the **computer server**.
- ❑ It is easy to connect a **computer or peripheral device**.
- ❑ The cable requirements are **relatively small**, resulting in **lower costs**.
- ❑ One of the disadvantages is that if the **main cable breaks**, the entire **network goes down**.
- ❑ This type of network is also **difficult to troubleshoot**.
- ❑ For these reasons, this type of topology is **not used** for large networks.



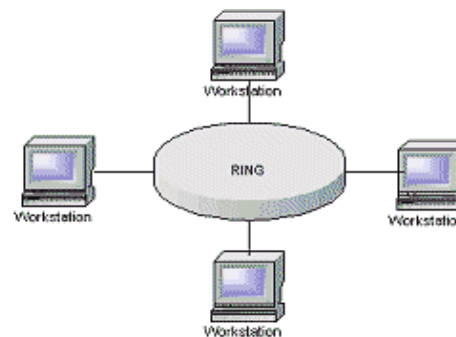
# Star Topology

- ❑ Each computer is connected to a central hub using a **point-to-point connection**.
- ❑ The **central hub** can be a computer server that manages the network.
- ❑ It is very popular because the **startup costs are low**.
- ❑ It is also easy to add new **nodes to the network**.
- ❑ The network is robust in the sense that if one connection between a computer and the hub fails, the other **connections remain intact**.
- ❑ If the **central hub fails**, however, the entire **network goes down**.
- ❑ It also requires more cable than **bus topology** and is, therefore, more expensive.



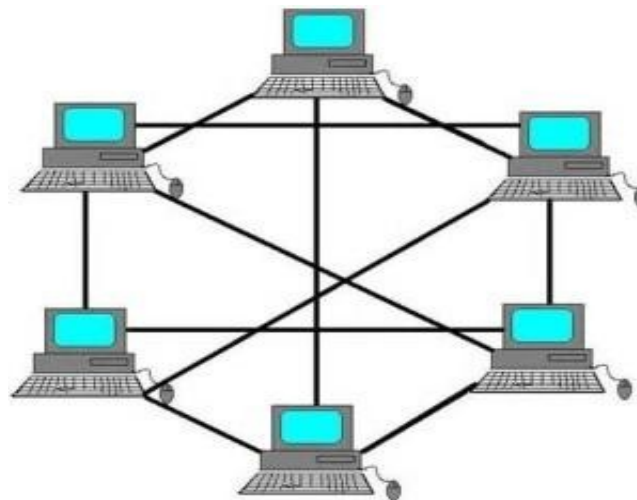
# Ring Topology

- ❑ The computers in the network are connected in a **circular fashion**, and the data travels in one direction.
- ❑ This type of network is easy to **install** and **manage**.
- ❑ If there is a problem in the network, it is **easy to pinpoint** which connection is **defective**.
- ❑ It is also good for handling **high-volume traffic** over long distances since every computer can act as a **booster of the signal**.
- ❑ On the downside, adding computers to this type of network is more **cumbersome**, and if one **single computer** fails, the entire network goes down.



# Mesh Topology

- In a mesh topology, every node has a **direct point-to-point connection** to every other node.
- Because all connections are direct, the network can handle **very high-volume traffic**.
- It is also **robust** because if one connection fails, the **others remain intact**.
- Security is also high since **data travels along a dedicated connection**.



# Transmission Media

- Two main categories:

  - Guided — wires, cables

  - Unguided — wireless transmission, e.g. radio, microwave, infrared, sound, sonar

- We will concentrate on **guided media** here:

  - Twisted-Pair cables:

    - Unshielded Twisted-Pair (UTP) cables

    - Shielded Twisted-Pair (STP) cables

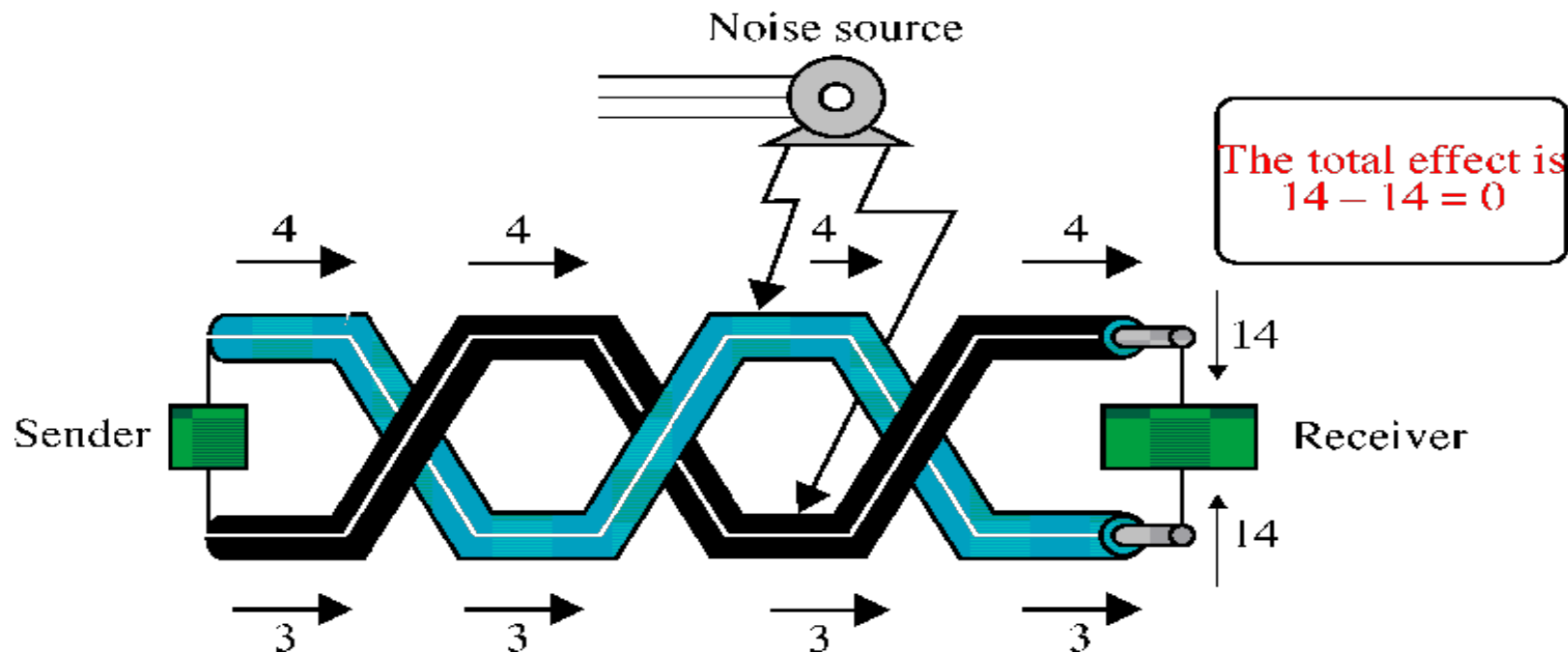
  - Coaxial cables

  - Fiber-optic cables

# Transmission Media (Cont.)

## Twisted-Pair Cables

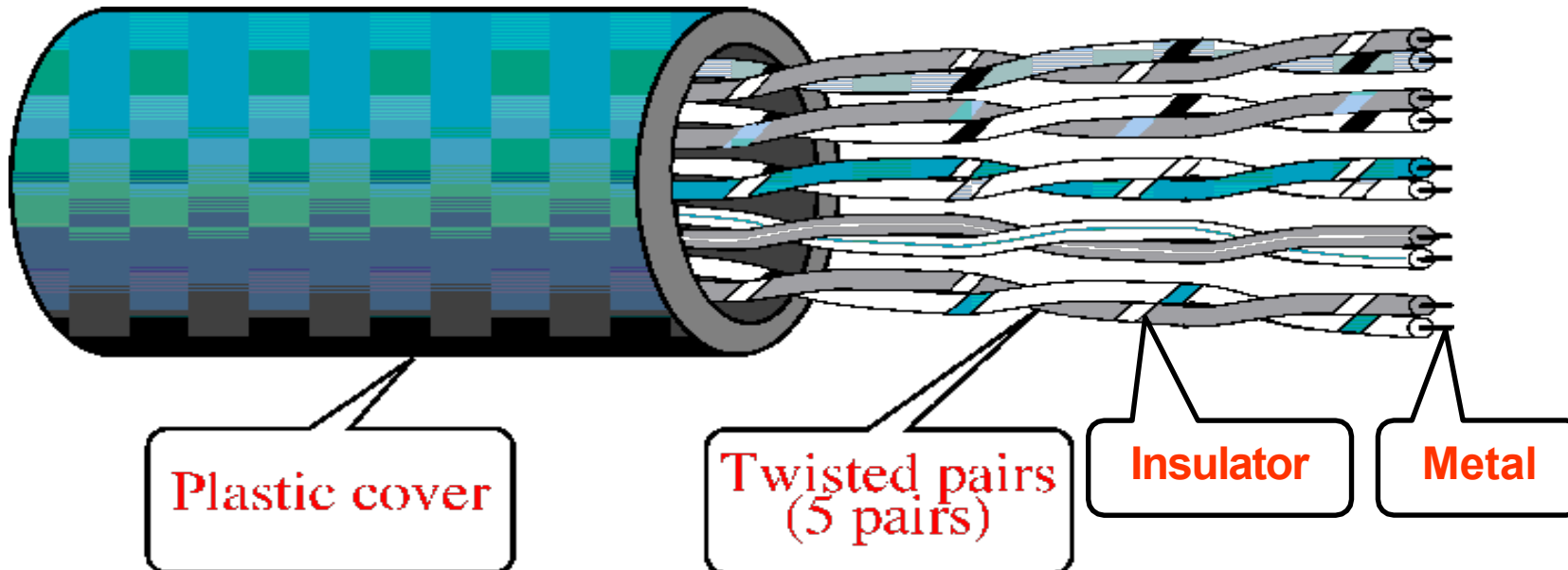
- If the pair of wires are not twisted, **electromagnetic noises** from, e.g., motors, will affect the closer wire more than the further one, thereby causing errors



# Transmission Media (cont.)

## Unshielded Twisted-Pair (UTP)

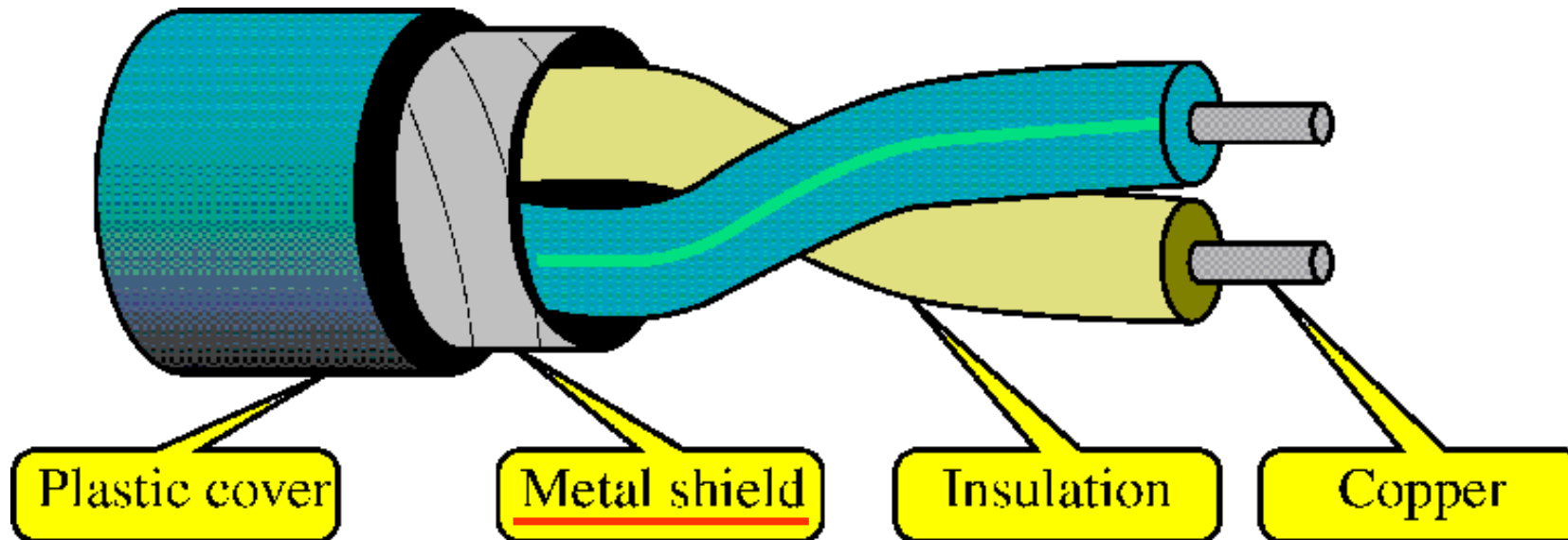
- Typically wrapped inside a **plastic cover** (for mechanical protection)
- A sample UTP cable with **5 unshielded twisted pairs of wires**



# Transmission Media (Cont.)

## Shielded Twisted-Pair (STP)

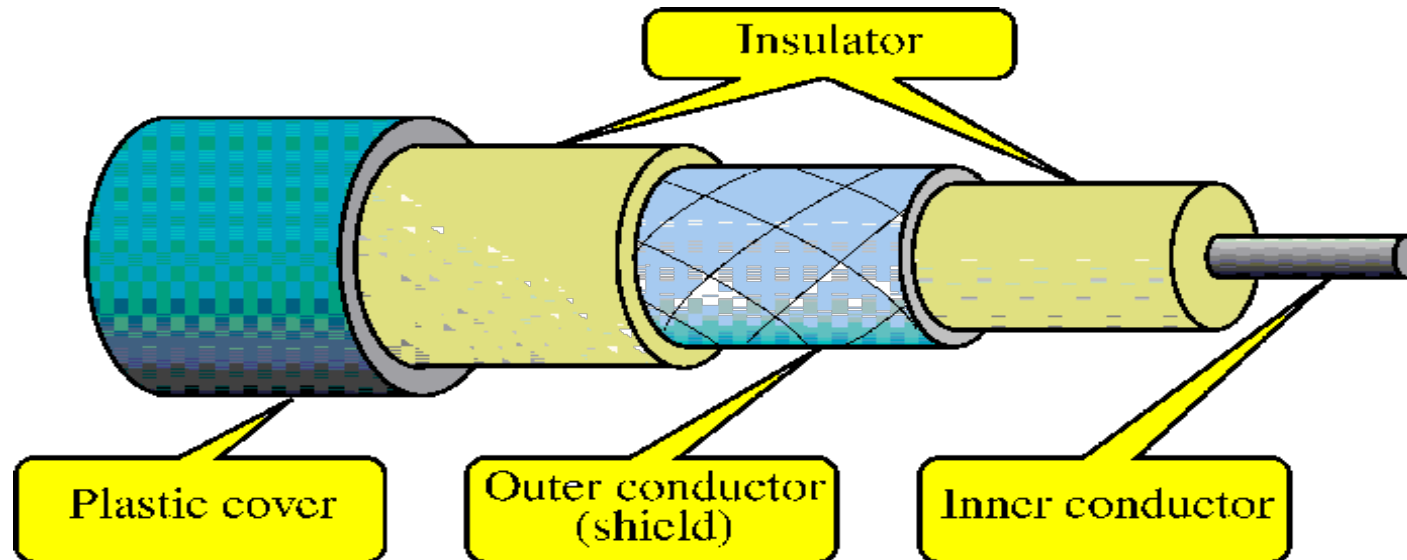
- STP cables are similar to UTP cables, except there is a metal foil or braided-metal-mesh cover that encases each pair of insulated wires



# Transmission Media (Cont.)

## Coaxial Cables

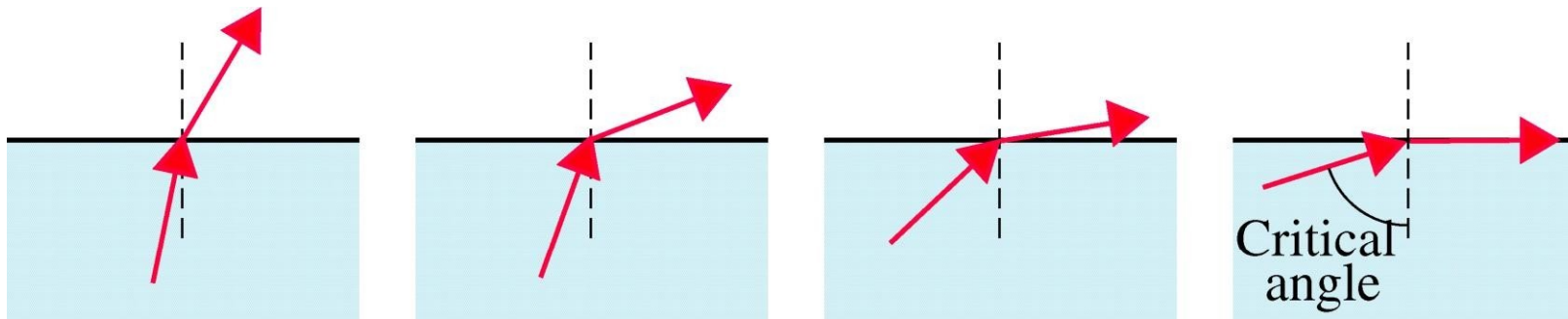
- In general, coaxial cables, or coax, carry signals of higher freq (100KHz–500MHz) than UTP cables
- Outer metallic wrapping serves both as a shield against noise and as the second conductor that completes the circuit



# Transmission Media (Cont.)

## Fiber-Optic Cables

- Light travels at  $3 \times 10^8 \text{ ms}^{-1}$  in free space and is the fastest possible speed in the Universe
- Light slows down in denser media, e.g. glass
- Refraction occurs at interface, with light bending away from the normal when it enters a less dense medium



- Beyond the critical angle  $\Rightarrow$  total internal reflection

# Transmission Media (Cont.)

## Fiber-Optic Cables

- An optical fiber consists of a core (denser material) and a cladding (less dense material)
- Simplest one is a multimode step-index optical fiber
- Multimode = multiple paths, whereas step-index = refractive index follows a step-function profile (i.e. an abrupt change of refractive index between the core and the cladding)
- Light bounces back and forth along the core
- Common light sources: LEDs and lasers

# Transmission Media (Cont.)

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# Transmission Media (Cont.)

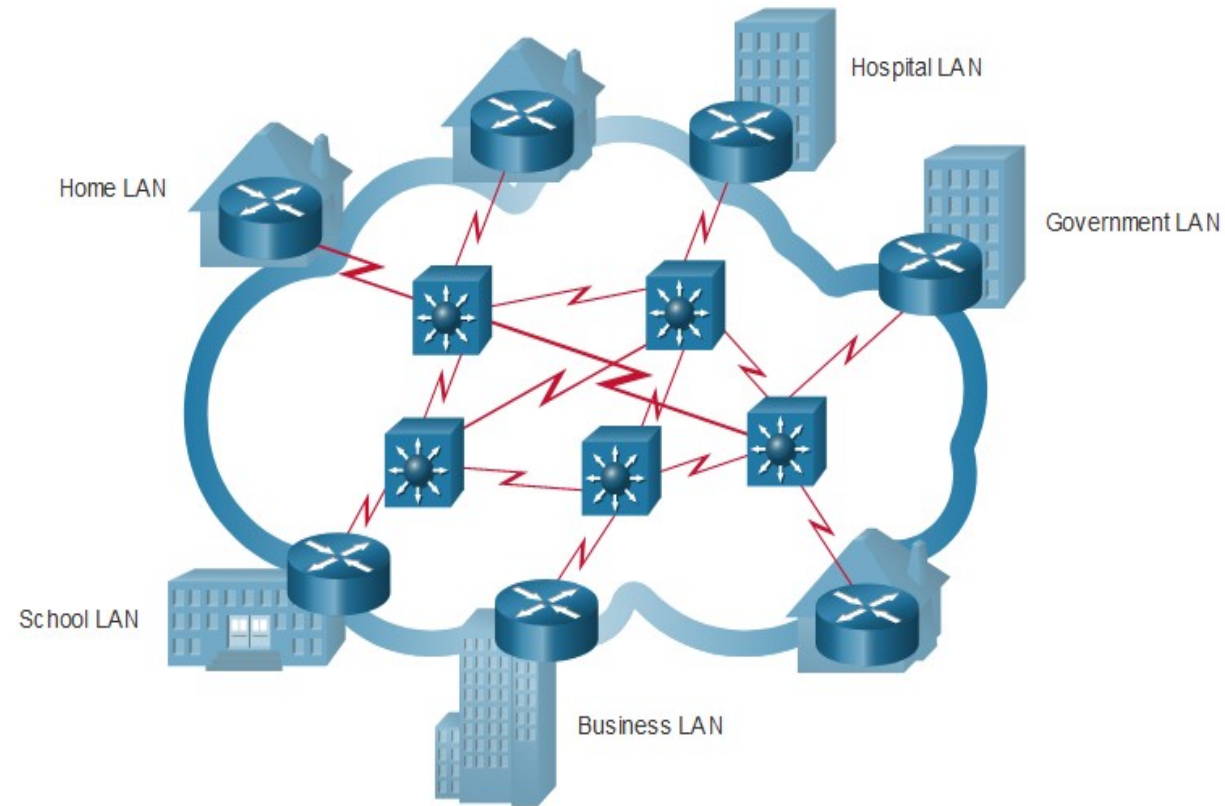
## Unbounded/Unguided Transmission Media

- Unguided or wireless media sends the data through the air (or water), which is available to anyone who has a device capable of receiving them.
- Types of unguided/ unbounded media are
  - Radio Transmission
  - Microwave Transmission

# The Internet

The internet is a worldwide collection of interconnected LANs and WANs.

- LANs are connected to each other using WANs.
- WANs may use copper wires, fiber optic cables, and wireless transmissions.
- The internet is not owned by any individual or group.
- The following groups were developed to help maintain structure on the internet:
  - ❑ IETF
  - ❑ ICANN
  - ❑ IAB



# The Internet-Based Service

Some of the basic services available to Internet users are –

- **Email** – a fast, easy, and inexpensive way to communicate with other Internet users around the world.
- **Telnet** – allows a user to log into a remote computer as though it were a local system.
- **FTP** – allows a user to transfer virtually every kind of file that can be stored on a computer from one Internet-connected computer to another.
- **World Wide Web (WWW)** – A hypertext interface to Internet information resources.

# Limitation of Internet

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- Theft of Personal information
- Negative effects on family communication
- Internet addiction
- Children using the Internet
- Virus threat
- Spamming

# The Web Concept

- Each page available on the website is called a web page.
- The first page of any website is called the home page for that site.

## What is WWW?

- WWW stands for World Wide Web.
- A technical definition of the World Wide Web is – All the resources and users on the Internet that are using the Hypertext Transfer Protocol HTTP.
- *The World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources*

# The Web Concept (Cont.)

## What is HTTP?

- HTTP stands for Hypertext Transfer Protocol.
- This is the protocol being used to transfer hypertext documents that make the World Wide Web possible.
- A standard web address such as [Yahoo.com](#) is called a URL and here the prefix HTTP indicates its protocol.

## What is a URL?

- URL stands for Uniform Resource Locator.
- Used to specify addresses on the World Wide Web.
- A URL is the fundamental network identification for any resource connected to the web.

# The Web Concept (Cont.)

A URL will have the following format

```
protocol://hostname/other_information
```

- The protocol specifies how information is transferred from a link.
- The protocol used for web resources is **HyperText Transfer Protocol (HTTP)**
- Other protocols compatible with most web browsers include **FTP**, **telnet**, **newsgroups**, and **Gopher**.
- The protocol is followed by a colon, two slashes, and then the domain name.
- The domain name is the computer on which **the resource is located**.
- **Links to particular files** or subdirectories may be further specified after the domain name.
- The directory names are **separated by single forward slashes**.

# The Web Concept (Cont.)

## What is a Web Server?

- Every Website sits on a computer known as a **Web Server**.
- This server is always connected to the **Internet**.
- Every Web server that is connected to the Internet is given a unique address made up of a series of four numbers between **0** and **256** separated by periods (IP address).
- For example, 68.178.157.132 or 68.122.35.127.

## What is a Web Browser?

- Web Browsers are **software installed** on your PC.
- To access the Web you need web browsers, such as Netscape Navigator, Microsoft Internet Explorer, or Mozilla Firefox.

# The Web Concept (Cont.)

## What is SMTP Server?

- SMTP stands for Simple Mail Transfer Protocol Server.
- This server takes care of delivering emails from one server to another server.
- When you send an email to an email address, it is delivered to its recipient by an SMTP Server.

## What is ISP?

- ISP stands for Internet Service Provider.
- They are the companies who provide you service in terms of internet connection to connect to the internet.
- You will buy space on a Web Server from any Internet Service Provider.
- This space will be used to host your Website.